higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

COMPUTER PROGRAMMING
(First Paper)
NQF LEVEL 4

15 MARCH 2019

This marking guideline consists of 6 pages.

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SECTION A

QUESTION 1

1.1	1.1.1	B		
	1.1.2	A		
	1.1.3	C		
	1.1.4	C		
	1.1.5	A		
	1.1.6	D		
	1.1.7	C		
	1.1.8	D		
	1.1.9	B		
	1.1.10	A		
	1.1.11	D		
	1.1.12	D		
	1.1.13	C		
	1.1.14	C		
	1.1.15	B		
			(15 × 1)	(15)
1.2	1.2.1	True		
	1.2.2	False		
	1.2.3	True		
	1.2.4	False		
	1.2.5	True		
			(5 × 1)	(5)
1.3	1.3.1	Multimedia		
	1.3.2	HTML		
	1.3.3	Sessionless / Stateless		
	1.3.4	OLE (object linking and embedding)		
	1.3.5	OOP		
			(5 × 1)	(5)
				[25]

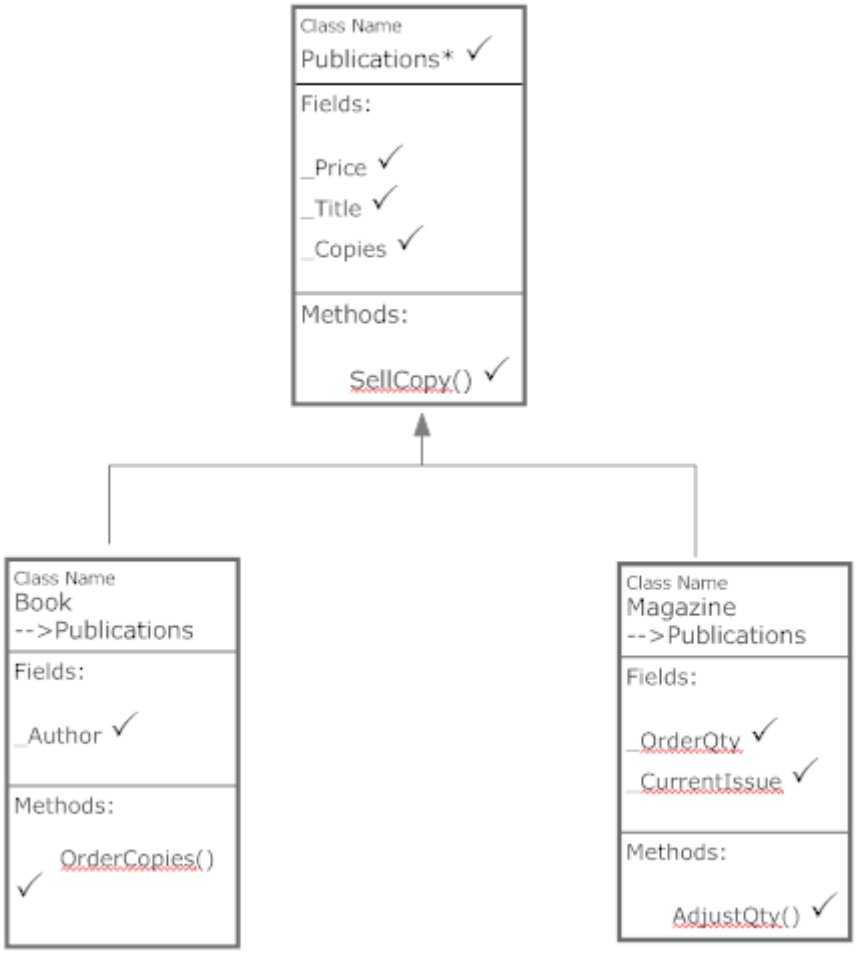
TOTAL SECTION A: 25

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SECTION B

QUESTION 2

2.1



(superclass does not have to be called “Publications”)

(10)

2.2 Minimum requirements:

- Microsoft IIS (latest version), Apache Server, XAMP or WAMP in Windows
- 2 × 1,6 GHz CPU or greater
- 4GB of RAM or greater
- 500Gb HDD or SDD or greater

(3)

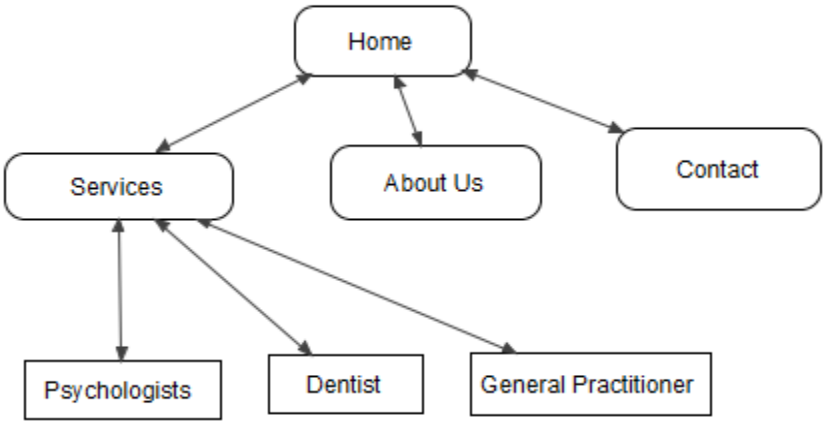
2.3 A storyboard is used during the initial stages as a way of communicating ideas ✓between the user and the developers✓, while a wireframe is the depiction of arrangements of a website's content and navigation✓ without colours or graphics because the main purpose is to show functionality✓. (4)

2.4
- reduce multimedia content in your pages
- limit client-server data transactions
- deploy JSON or similar data technology
- limit interactive content on web pages
- produce website as a one page site with click to scroll HTML5 options
- Make less interactive webpages.
- Or any other relevant response (2)

[19]

QUESTION 3

3.1



✓ each for "Services", "Psychologists", "Dentists", "General Practitioners" correct level
✓ for the rest i.e. Home, About US & Contact

Or any other logical representation (4 + 1) (5)

3.2
- Takes longer to learn how to use
- Does not support SEO
- Produces unnecessary and complicated code
- Or any other acceptable disadvantage (3)

3.3 3.3.1
```
INSERT INTO EMPLOYEE✓
(EmpID, LastName, CellPhone) ✓
VALUES (104, 'Zulu', '072 664 0025') ✓
```
(3)

3.3.2
```
UPDATE EMPLOYEE✓
SET LastName = 'Tempest', CellPhone = '082 100 3999'✓
WHERE EmpID = 100✓
```
(3)

3.3.3
```
DELETE FROM EMPLOYEE✓
WHERE EmpID = 102✓
```
(2)

3.4 3.4.1 When the web browser loads the page (1)

3.4.2 It will show a dialogue box with the user's calculated age (1)

3.4.3 Declare and initialise the CurrentYear✓
var CurrentYear=(new Date()).getFullYear() / var CurrentYear = currentTime.getFullYear()✓ (2)
[20]

QUESTION 4

4.1

COHESION	COUPLING
Refers to the relationship within a class✓	Refers to the relationship between two or more classes✓
Degree to which the methods within a class focus on a single task✓	Degree to which the methods from different classes are connected to one another✓
Designers strive for high cohesion✓	Designers strive for low coupling✓

(3 + 3) (6)

4.2
- HTML/5✓, CSS✓, Ajax✓, JavaScript✓, JQuery✓, Active Server Pages✓, MooTools✓, Dojo Toolkit✓ (or any other recognised and relevant software technology) (3)

4.3 4.3.1 <script src='script.js'>/</script>✓✓ (2)

4.3.2 <h2>ABSA Premier League</h2> (1)

4.3.3 <th>/<th> (1)

4.3.4 (a) It will display a hyperlink to the "goals.html" file
(b)
(2 × 1) (2)

4.4 Ownership means the software legally belongs to whoever owns the copyright for the software and the owner of the PC has the right to use the software, while copyright means whoever commissioned the creation of the software owns the copyright, but the buyer or developer do not automatically own the software. (2 + 2) (4)
[19]

QUESTION 5

5.1 5.1.1 Process of connecting a data source to end-user interface controls

5.1.2 Subset of the database stored on RAM for faster access

5.1.3 Software object that connects the dataset and database **OR**
It can be a memory-resident representation of data that provides a consistent relational programming model independent of the data source"

(3 × 1) (3)

5.2
- http:// – tells the web browser which protocol to use
- tvet.edu.ac.za – domain name of the web server
- notes – name of the folder or directory
- cssnotes – name of the file of the webpage to be viewed (4)

5.3 **Table 1**: Students details✓

Student No# (PK)✓
Student Name
Phone No#

Table 2: Assessments types✓

Ass No# (PK)✓
Ass Name

Table 3: Assessment marks✓

Student No# (FK)	(PK)✓
Ass No# (FK)	(PK)
Mark	Mark obtained

(3 × 2) (6)

5.4 5.4.1
- The computer system requests a username and password and checks them against the ones stored on the system.
- It locks the user out after more than three incorrect attempts.

5.4.2
- The computer system generates a new OTP and sends it through e-mail or SMS.
- It is valid for one login session

(2 × 2) (4)
[17]

TOTAL SECTION B: 75
GRAND TOTAL: 100

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